

Chapter Three

Crystal Binding

Main interaction responsible to hold the crystals together are the attractive electrostatic interaction between the negative charge of electrons and positive charge of nucleus are most important interaction. Other forces such as magnetic forces are weak force and gravitational forces are negligible effect.

The cohesive energy of a crystal is defined as the energy that must be added to the crystal to separate its components into neutral free atom at rest, at infinite separation with electronic configuration. The term lattice energy is used in the discussion of ionic crystal and is defined as energy that must be added to crystal to separate its component ions free ions at the rest at infinite separation.

3.1 Crystal of Inert Gases

The inert gases form simplest crystals. The electron distribution is very close to the free atoms. They are transparent insulator, weakly bound, with low melting temperatures. The atoms have high ionization energies.

The outer most electron shell of the atom are completely filled and distribution of electron charge in free atom is spherically symmetric. The electron distribution in the crystal cannot be significantly distorted from the electron distribution around the free atoms, because cohesive energy is less than ionization energy.

Van der Waal - London Interaction.

Consider two identical inert gas atoms at separation R comparable with radii of atoms. If the charge distribution on the atom were rigid, the interaction between atoms would be zero, because the electrostatic potential of spherical distribution of electronic charge cancelled outside a neutral atoms by the electrostatic potential of the charge on nucleus. Then inert gas atom show no cohesion and could not condense. But the atoms induce dipole moment in each other, and the induced moments cause an attractive interaction between atoms.



Figure 3.1: Coordinates of the two oscillators

As a model we consider two identical linear harmonic oscillator 1 and 2 separated by R each atom experience charges $\pm e$ with separation x_1 and x_2 , the particle oscillate along x axis. Let

p_1 and p_2 denote momenta and c is force constant.

$$H_0 = \frac{1}{2m} p_1^2 + \frac{1}{2} c x_1^2 + \frac{1}{2m} p_2^2 + \frac{1}{2} c x_2^2$$

Assume that $c = m\omega_0^2$

$$\text{Let the } H_1 = \frac{e^2}{R} + \frac{e^2}{R+x_1-x_2} - \frac{e^2}{R+x_1} - \frac{e^2}{R-x_2}$$

$$|x_1|, |x_2| \ll R \text{ we expand } H_1 \simeq -\frac{2e^2 x_1 \cdot x_2}{R^3}$$

$$x_s = \frac{1}{\sqrt{2}}(x_1 + x_2) \quad x_a = \frac{1}{\sqrt{2}}(x_1 - x_2)$$

$$x_1 = \frac{1}{\sqrt{2}}(x_s + x_a) \quad x_2 = \frac{1}{\sqrt{2}}(x_s - x_a)$$

$$p_1 = \frac{1}{\sqrt{2}}(p_s + p_a) \quad p_2 = \frac{1}{\sqrt{2}}(p_s - p_a)$$

$$H = \left[\frac{1}{2m} p_s^2 + \frac{1}{2} \left(c - \frac{2e^2}{R} \right) x_s^2 \right] + \left[\frac{1}{2m} p_a^2 + \frac{1}{2} \left(c - \frac{2e^2}{R} \right) x_a^2 \right]$$

The two frequency of coupled equation is given by

$$\omega = \left[\frac{\left(c \pm \frac{2e^2}{R^3} \right)}{m} \right]^{1/2} = \omega_0 \left[1 \pm \frac{1}{2} \left(\frac{2e^2}{cR^3} \right) - \frac{1}{8} \left(\frac{2e^2}{cR^3} \right)^2 \dots \right] \text{ with } \omega_0 = \left(\frac{c}{m} \right)^{1/2}$$

The zero-point energy is given as $\frac{1}{2} \hbar (\omega_a + \omega_s)$ because of the interaction the sum is lowered

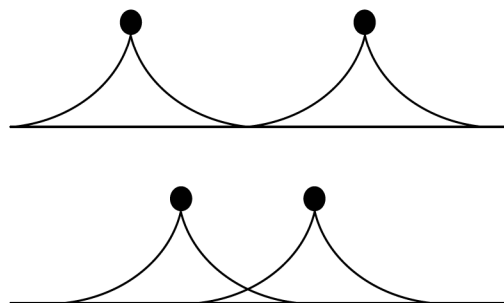
from the uncoupled value $2 \cdot \frac{1}{2} \hbar \omega_0$ by $\frac{1}{2} \hbar (\Delta \omega_s + \Delta \omega_a) = -\hbar \omega_0 \cdot \frac{1}{8} \left(\frac{2e^2}{cR^3} \right)^2 = -\frac{A}{R^6}$

The attractive interaction varies as the minus sixth power of the separation of two oscillators.

This is identified as Vander Waals interaction, it is also identified as London interaction or induced dipole -dipole interaction.

Repulsive term

As the two atoms are brought together their charge distribution gradually overlap, thereby changing the electrostatic energy of system. At sufficiently close separation the overlap energy is repulsive in large part because of Pauli exclusion principle. In the figure Electronic



charge distributions overlap as atoms approach. The solid circles denote the nuclei The experimental data on inert gas be fitted well by an empirical repulsive potential of the form

$\frac{B}{R^{12}}$, where B is positive constant.

Total potential energy and Lennard-Jones potential

Potential energy of Diatomic molecule in term of interatomic distance is given by

$U(R) = -\frac{A}{R^6} + \frac{B}{R^{12}}$ where $\frac{A}{R^6}$ is long range attractive potential and $\frac{B}{R^{12}}$ is short range repulsive potential.

Using data of Virial coefficient and viscosity new parameter ϵ and σ is introduced where $A = 4\epsilon\sigma^6$ and $B = 4\epsilon\sigma^{12}$ the Lennard-Jones potential is given by

$$U(R) = 4\epsilon \left[\left(\frac{\sigma}{R} \right)^{12} - \left(\frac{\sigma}{R} \right)^6 \right]$$

The force between two atom is given by $-\frac{dU}{dR}$

The value of σ and ϵ can be experimentally determined.

Generalization of theory

If Potential energy of Diatomic molecule is given by $U(R) = -\frac{A}{R^n} + \frac{B}{R^m}$ then force is given by

$$F(R) = -\frac{dU}{dR} \Big|_{R_0} \text{ which is equivalent to } F = -\left(\frac{nA}{R^{n+1}} - \frac{mB}{R^{m+1}} \right) = -\frac{nA}{R^{n+1}} + \frac{mB}{R^{m+1}}$$

For equilibrium separation at $R = R_0$,

$$F = 0 \Rightarrow \frac{nA}{R_0^{n+1}} - \frac{mB}{R_0^{m+1}} = 0 \Rightarrow \frac{nA}{R_0^{n+1}} = \frac{mB}{R_0^{m+1}} \Rightarrow R_0 = \left(\frac{nA}{mB} \right)^{\frac{1}{n-m}} \Rightarrow \frac{B}{A} \frac{R_0^n}{R_0^m} = \frac{n}{m}$$

The potential energy at equilibrium separation is given by

$$U(R_0) = -\frac{A}{R_0^n} + \frac{B}{R_0^m} \Rightarrow -\frac{A}{R_0^n} \left(1 - \frac{B}{A} \frac{R_0^n}{R_0^m} \right)$$

$$\text{Where } U(R_0) = -\frac{A}{R_0^n} \left(1 - \frac{n}{m} \right)$$

$$R_0 \text{ is stable equilibrium point so } \frac{d^2U}{dR^2} \Big|_{R=R_0} > 0 \Rightarrow m > n$$

Example 1: The potential energy of a system of two atoms is given by

$$U(R) = -\frac{\alpha}{R^4} + \frac{\beta}{R^{12}} \quad (1)$$

Calculate the amount of energy released when the atoms form a stable bond. Determine the bond length.

Solution: We know that the force of interaction between the two atoms is given by

$$F = -\frac{dU}{dR} = -\frac{4\alpha}{R^5} + \frac{12\beta}{R^{13}}$$

However, at an equilibrium distance (= bond length) $R = R_0$, the net force acting on the atoms is zero. Therefore,

$$F = -\left(\frac{dU}{dR}\right)_{R=R_0} = 0 \Rightarrow \frac{4\alpha}{R_0^5} = \frac{12\beta}{R_0^{13}} \Rightarrow R_0^8 = \frac{12\beta}{4\alpha} = \frac{3\beta}{\alpha}, R_0 = \left(\frac{3\beta}{\alpha}\right)^{1/8}$$

This is the required bond length.

Now substituting the value of R_0 in expression of potential energy.

Obtain the amount of energy released, i.e. $U_e = \frac{\alpha}{R_0^4} \left(1 - \frac{1}{3}\right) = \frac{\alpha}{\left(\frac{3\beta}{\alpha}\right)^{1/2}} \times \frac{2}{3} = -\left(\frac{4\alpha^3}{27\beta}\right)^{1/2}$

This is the required energy when the atoms form a stable bond.

Example 2: The potential energy of a system is given as $U(R) = -\frac{A}{R^6} + \frac{B}{R^{12}}$, the atoms form a stable bond length of $3A^0$ and bond energy of $1.8eV$. Calculate the force required to break the molecule and the critical interatomic distance for which it occurs.

Solution: $R_0 = 3A^0 = 3 \times 10^{-10} m$, $D = -U_0 = 1.8eV = 1.8 \times 1.6 \times 10^{-19} J$, $m = 12, n = 6$

$$D = -U = \frac{A}{R_0^n} \left(1 - \frac{n}{m}\right) \Rightarrow D = 2DR_0^6 = 2 \times 1.8 \times 1.6 \times 10^{-19} \times (3 \times 10^{-10})^6 = 4.19 \times 10^{-76} Jm^6$$

$$F = -\left(\frac{dU}{dR}\right)_{R=R_0} = -\frac{6A}{R_0^7} + \frac{12B}{R_0^{13}} = 0 \Rightarrow B = \frac{A(R_0)^6}{2} = \frac{A(R_0)^6}{2}$$

$$= \frac{4.19 \times 10^{-76} \times (3 \times 10^{-10})^6}{2} = 1.53 \times 10^{-133} Jm^{12}$$

The critical separation is given by $\frac{dF}{dR} = 0 \Rightarrow \frac{42A}{R_c^8} - \frac{156B}{R_c^{14}} = 0 \Rightarrow R_c^6 = \frac{13}{7} R_0^6 \Rightarrow R_c = \left(\frac{13}{7}\right)^{1/6} R_0$

$R_c = 3.3A^0$

Minimum force, $F_{\min} = -\left(\frac{dU}{dR}\right)_{R=R_c} = -\frac{6A}{R_c^7} + \frac{12B}{R_c^{13}} = -\frac{6A}{R_c^7} \left(1 - \frac{2B}{AR_c^6}\right) = -\frac{6A}{R_c^7} \left[1 - \left(\frac{R_0}{R_c}\right)^6\right]$

Where $\frac{B}{A} = \frac{R_0^6}{2}$ so $F_{\min} = -\frac{6 \times 4.19 \times 10^{-76}}{(3.33 \times 10^{-10})^7} \left[1 - \left(\frac{3}{3.33}\right)^6\right] = -2.58 \times 10^{-9} N$

Thus the force needed to dissociate the molecule is $2.58 \times 10^{-9} N$

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3.2 Ionic Crystals and Electrostatic or Madelung Energy

Ionic crystals are made up of positive and negative ions. The **ionic bond** results from the electrostatic interaction of oppositely charged ions. Two common crystal structure structures found in ionic crystals, the sodium chloride and the cesium chloride.

Madelung energy The long range interaction between ions with $\pm q$ is the electrostatic interaction $\pm \frac{q^2}{r}$, attractive between ions with opposite charge and repulsive between ions of same charge. The repulsive interaction between ions are same as inert gas atom but Vander Waal part of interaction has relatively small contribution to the cohesive interaction in ionic crystal is order of 1 or 2 percent. The main contribution to the binding energy of ionic crystals is electrostatic and is called the Madelung energy.

If U_{ij} is the interaction energy between ions i and j , we define a sum U_i which includes all interaction involving the ion i :

$$U_i = \sum_j U_{ij}, \text{ where the summation includes all ions except } j = i.$$

we suppose that U_{ij} may be written as sum of a central field repulsive potential of form $\lambda \exp(-r/\rho)$, where λ and ρ are empirical parameters, and a coulomb potential $\pm \frac{q^2}{r}$.

Thus, in CGS unit $U_{ij} = \lambda \exp\left(-\frac{r_{ij}}{\rho}\right) \pm \frac{q^2}{r_{ij}}$, where the positive sign is taken for like charges and negative sign is for unlike charges. In SI units the coulomb interaction is $\pm \frac{q^2}{4\pi\epsilon_0 r_{ij}}$.

It is convenient to introduce quantities p_{ij} such that $r_{ij} = p_{ij}R$ where R is the nearest-neighbour separation in the crystal. If we include repulsive interaction only among nearest neighbours, we have

$$U_{ij} = \begin{cases} \lambda \exp\left(-\frac{R}{\rho}\right) - \frac{q^2}{R} & (\text{nearest neighbors}) \\ \pm \frac{1}{p_{ij}} \frac{q^2}{R} & (\text{otherwise}) \end{cases}$$

$U_{total} = NU_i = N\left(z\lambda \exp\left(-\frac{R}{\rho}\right) - \frac{\alpha q^2}{R}\right)$, where z is the number of nearest neighbours of any ion and $\alpha = \sum_j \frac{(\pm)}{p_{ij}}$ is identified as Madelung constant.

For simplicity we are taking $U = N\left[-\frac{\alpha q^2}{4\pi\epsilon_0 R} + B \exp\left(-\frac{R}{\rho}\right)\right]$

Madelung energy

$$U_{total} = NU_i = N\left(z\lambda \exp\left(-\frac{R}{\rho}\right) - \frac{\alpha q^2}{R}\right)$$

$$N \frac{dU_i}{dR} = -\frac{Nz\lambda}{\rho} \exp\left(-\frac{R}{\rho}\right) + \frac{N\alpha q^2}{R^2} = 0$$

$$R_0^2 \exp\left(-\frac{R_0}{\rho}\right) = \rho \alpha \frac{q^2}{z\lambda} \text{ to determine equilibrium separation } R_0 \text{ if the parameters } \rho, \lambda \text{ of}$$

repulsive interaction are known. For SI unit replace q^2 by $\frac{q^2}{4\pi\epsilon_0}$.

The total lattice energy of crystal $2N$ ions at their equilibrium separation R_0 may be written as

$$U_{tot} = -\frac{N\alpha q^2}{R_0} \left(1 - \frac{\rho}{R_0}\right). \text{ The term } -\frac{N\alpha q^2}{R_0} \text{ is identified as Madelung energy. We can find that}$$

ρ is of order of $0.1R_0$. so that repulsive interaction has very short range. In some place

Madelung constant is also represented A . In SI unit $U_{tot} = -\frac{N\alpha q^2}{4\pi\epsilon_0 R_0} \left(1 - \frac{\rho}{R_0}\right) kJ / kmole$. Where

N is Avogadro number

Evaluation of the Madelung constant

The Madelung constant is given by $\alpha = \sum_j \frac{(\pm)}{p_{ij}}$ if we take reference ion as a negative charge the

plus sign will apply to positive ions and minus sign to negative ions the equivalent definition is

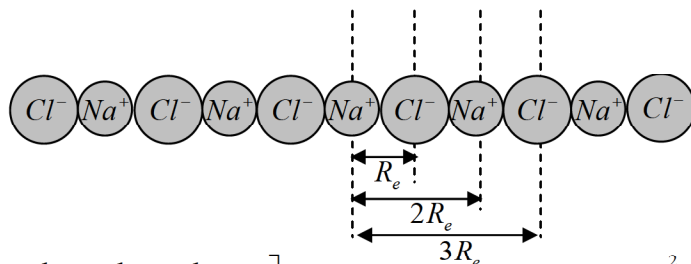
$\frac{\alpha}{R} = \sum_j \frac{\pm}{r_j}$ where r_j is j^{th} ions from the reference ion and R is the nearest neighbour distance

It must be emphasized that the value given for α will depend on whether it is defined in term of the largest neighbour distance R or in terms of the lattice parameter a or in term of relevant length.

Example: Show that the Madelung constant α for an infinite linear chain of ions of alternating charge at an equilibrium separation is $2 \ln 2$.

Solution: Madelung constant A is a function of crystal structure and can therefore be calculated from the geometrical arrangement of ions in a crystal. In the present case, we have to calculate the Madelung constant for an infinite linear chain of ions of alternating charge (i.e. a one dimensional $NaCl$ crystal) at equilibrium separation R_0 . Let us take one sodium ion as the reference ion and then write various terms of the series for alternating ions in terms of R_0 ,

we obtain



$$\frac{\alpha}{R_0} = 2 \left[\frac{1}{R_0} - \frac{1}{2R_0} + \frac{1}{3R_0} - \frac{1}{4R_0} + \dots \right] \quad \text{We know that } \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\Rightarrow \alpha = 2 \left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \right] \Rightarrow \alpha = 2 \ln 2$$

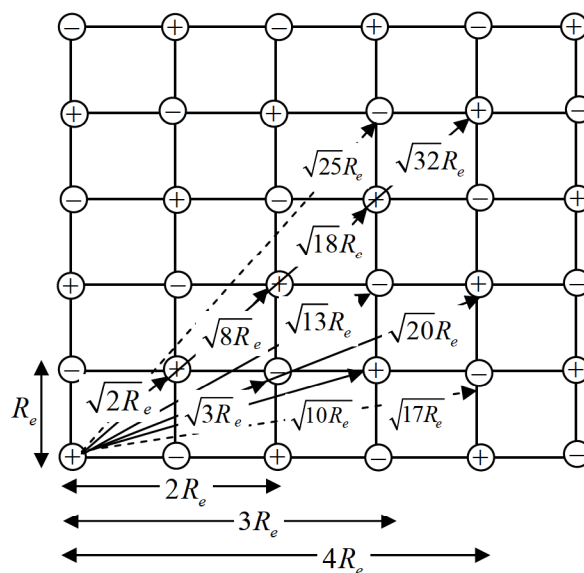
The factor 2 occurs because there are two ions, one to the right and one to the left.

Example: Calculate the Madelung constant for an infinite array of two-dimensional ionic charge distribution (i.e. two dimensional $NaCl$ crystal) with equilibrium ion separation as R_0 .

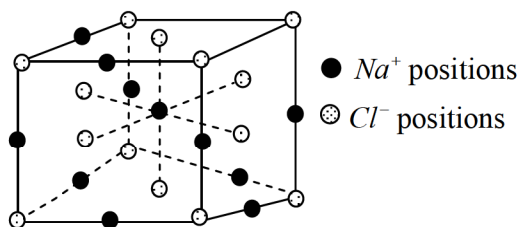
Solution: Consider a two dimensional array of ionic charge (alternate Na^+ and Cl^- ion) distribution as shown as figure. Let us take Na^+ ion (at left bottom corner) as the reference ion and then write various term of the series for alternating ions in term increasing value of R_e , we have

$$A = 4 \left[\left(\frac{1}{1} - \frac{1}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{4}} - \frac{2}{\sqrt{5}} - \frac{1}{\sqrt{8}} \right) + \left(\frac{1}{3} - \frac{2}{\sqrt{10}} + \frac{2}{\sqrt{13}} - \frac{1}{\sqrt{18}} \right) + \left(-\frac{1}{\sqrt{16}} + \frac{2}{17} - \frac{2}{\sqrt{20}} + \frac{2}{\sqrt{25}} - \frac{1}{\sqrt{32}} \right) + \dots \right]$$

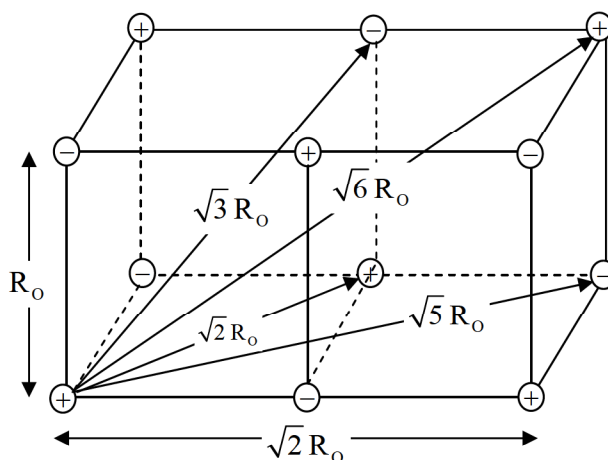
$$= 1.1716 + 0.1635 + 0.0795 + 0.0443 + \dots \approx 1.6135$$



Example: The three dimensional crystal for NaCl is given in figure find the Madelung constant



Solution:



$$\alpha = \frac{6}{1} - \frac{12}{\sqrt{2}} + \frac{8}{\sqrt{3}} - \frac{6}{\sqrt{4}} + \frac{24}{\sqrt{5}} - \frac{24}{\sqrt{6}} + \dots$$

Which will converge to value 1.74

3.3 Bulk Modulus and Compressibility

The bulk modulus is given by $\beta = -V \left(\frac{dP}{dV} \right)$ and compressibility is defined as $\frac{1}{K} = \beta = -V \left(\frac{dP}{dV} \right)$

From first law of thermodynamics $dQ = dU + dW$ at absolute zero $dQ = 0$

So $\frac{dU}{dV} = -P$, $\frac{d^2U}{dV^2} = -\frac{dP}{dV}$ so $\frac{1}{k_0} = V_0 \left(\frac{d^2U}{dV^2} \right)$ where V_0 is volume of crystal at $R = R_0$

For NaCl crystal $R = R_{cl} + R_{Na} = \frac{a}{2}$ and volume per molecule is $= \frac{a^3}{4}$

Therefore $V = \frac{a^3}{4} N = 2NR^3$ so $\frac{dV}{dR} = 6NR^2 \Rightarrow \frac{dR}{dV} = \frac{1}{6NR^2}$

$$\frac{dU}{dV} = \left(\frac{dU}{dR} \right) \cdot \left(\frac{dR}{dV} \right) \Rightarrow \frac{d}{dV} \left(\frac{dU}{dV} \right) = \left(\frac{dU}{dR} \right) \cdot \frac{d}{dV} \left(\frac{dR}{dV} \right) + \left(\frac{dR}{dV} \right) \cdot \frac{d}{dV} \left(\frac{dU}{dR} \right)$$

$$\left(\frac{d^2U}{dV^2} \right) = \left(\frac{dU}{dR} \right) \cdot \left(\frac{d^2R}{dV^2} \right) + \left(\frac{dR}{dV} \right) \frac{d}{dR} \left(\frac{dU}{dR} \right) \cdot \left(\frac{dR}{dV} \right)$$

$$\left(\frac{d^2U}{dV^2}\right) = \left(\frac{dU}{dR}\right) \cdot \left(\frac{d^2R}{dV^2}\right) + \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right)$$

At equilibrium separation

$$U = N \left[-\frac{\alpha q^2}{4\pi \epsilon_0 R} + B \exp\left(-\frac{R}{\rho}\right) \right] \text{ For } \left(\frac{dU}{dR}\right) = 0 \text{ and } V = V_0 = 2NR_0^3$$

$$\text{So } \left(\frac{d^2U}{dV^2}\right) = \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right) \text{ so } \frac{1}{K_0} = 2NR_0^3 \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right)$$

$$\frac{1}{K_0} = 2NR_0^3 \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right) \Rightarrow \frac{2R_0^3}{36R_0^4} \left(\frac{d^2U}{dR^2}\right) = \frac{1}{18NR_0} \left(\frac{d^2U}{dR^2}\right) = \frac{\alpha q^2 N}{R_0^3} \left(\frac{R_0}{\rho} - 2\right)$$

$$\frac{1}{K_0} = \beta = \frac{\alpha q^2}{18R_0^4} \left(\frac{R_0}{\rho} - 2\right) \Rightarrow \frac{R_0}{\rho} = \frac{18R_0^4 \beta}{\alpha q^2} + 2.$$

Bond dissociation energy of ionic molecule is given by

$DE = -U_{total} - IP + EA$. where U_{total} potential energy, IP is ionisation potential and EA is electron Affinity.

Example: if $\frac{B}{R^m}$ is repulsive term like $\frac{\rho}{R_0}$ then prove that value of $m = 1 + \frac{72\pi \epsilon_0 R_0^4}{K_0 \alpha q^2}$ and

$$U = -\frac{N\alpha q^2}{4\pi \epsilon_0 R_0} \left(1 - \frac{1}{m}\right)$$

Solution: $U(R) = -N \left[\frac{\alpha q^2}{4\pi \epsilon_0 R} + \frac{B}{R^m} \right] \quad V = V_0 = 2NR_0^3$

Hence, $\frac{dU}{dR} = 0$ at $R = R_0 \Rightarrow R_0^{m-1} = \frac{mB4\pi \epsilon_0}{\alpha q^2} \Rightarrow R_0 = \left(\frac{4\pi \epsilon_0 mB}{\alpha q^2}\right)^{\frac{1}{m-1}}$

$$\frac{1}{K_0} = 2NR_0^3 \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right), \quad \frac{dR}{dV} = \frac{1}{6NR^2} \quad \text{Where } \frac{d^2U}{dR^2} = N \left(-\frac{2\alpha q^2}{4\pi \epsilon_0 R_0^3} + \frac{m(m+1)}{R_0^{m+2}} \right)$$

$$\frac{1}{K_0} = 2NR_0^3 \left(\frac{1}{6NR^2}\right)^2 \cdot N \left(\frac{m(m+1)B}{R_0^{m+2}} - \frac{2\alpha q^2}{4\pi \epsilon_0 R_0^3} \right)$$

$$\frac{1}{K_0} = \frac{1}{18R_0^4} \cdot \left(\frac{m(m+1)B}{R_0^{m-2}} - \frac{2\alpha q^2}{4\pi \epsilon_0} \right) \Rightarrow \frac{1}{K_0} = \frac{1}{18R_0^4} \cdot \left(\frac{m(m+1)B\alpha q^2}{mB4\pi \epsilon_0} - \frac{2\alpha q^2}{4\pi \epsilon_0} \right)$$

$$\Rightarrow \frac{1}{K_0} = \frac{\alpha q^2}{72\pi \epsilon_0 R_0^4} [m+1-2] \Rightarrow m = 1 + \frac{72\pi \epsilon_0 R_0^4}{K_0 \alpha q^2}$$

Total energy, $U(R) = -N \left[\frac{\alpha q^2}{4\pi \epsilon_0 R} + \frac{B}{R^m} \right]$ put $R_0^{m-1} = \frac{mB4\pi \epsilon_0}{\alpha q^2}$ and $U = \frac{-\alpha q^2}{4\pi \epsilon_0 R_0} \left[1 - \frac{1}{m} \right]$

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$$\frac{dU}{dV} = \left(\frac{dU}{dR} \right) \cdot \left(\frac{dR}{dV} \right) \Rightarrow \frac{d}{dV} \left(\frac{dU}{dV} \right) = \left(\frac{dU}{dR} \right) \cdot \frac{d}{dV} \left(\frac{dR}{dV} \right) + \left(\frac{dR}{dV} \right) \cdot \frac{d}{dV} \left(\frac{dU}{dR} \right)$$

$$\left(\frac{d^2U}{dV^2} \right) = \left(\frac{dU}{dR} \right) \cdot \left(\frac{d^2R}{dV^2} \right) + \left(\frac{dR}{dV} \right) \frac{d}{dR} \left(\frac{dU}{dR} \right) \cdot \left(\frac{dR}{dV} \right)$$

$$\left(\frac{d^2U}{dV^2}\right) = \left(\frac{dU}{dR}\right) \cdot \left(\frac{d^2R}{dV^2}\right) + \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right)$$

At equilibrium separation

$$U = N \left[-\frac{\alpha q^2}{4\pi \epsilon_0 R} + B \exp\left(-\frac{R}{\rho}\right) \right] \text{ For } \left(\frac{dU}{dR}\right) = 0 \text{ and } V = V_0 = 2NR_0^3$$

$$\text{So } \left(\frac{d^2U}{dV^2}\right) = \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right) \text{ so } \frac{1}{K_0} = 2NR_0^3 \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right)$$

$$\frac{1}{K_0} = 2NR_0^3 \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right) \Rightarrow \frac{2R_0^3}{36R_0^4} \left(\frac{d^2U}{dR^2}\right) = \frac{1}{18NR_0} \left(\frac{d^2U}{dR^2}\right) = \frac{\alpha q^2 N}{R_0^3} \left(\frac{R_0}{\rho} - 2\right)$$

$$\frac{1}{K_0} = \beta = \frac{\alpha q^2}{18R_0^4} \left(\frac{R_0}{\rho} - 2\right) \Rightarrow \frac{R_0}{\rho} = \frac{18R_0^4 \beta}{\alpha q^2} + 2.$$

Bond dissociation energy of ionic molecule is given by

$DE = -U_{total} - IP + EA$. where U_{total} potential energy, IP is ionisation potential and EA is electron Affinity.

Example: if $\frac{B}{R^m}$ is repulsive term like $\frac{\rho}{R_0}$ then prove that value of $m = 1 + \frac{72\pi \epsilon_0 R_0^4}{K_0 \alpha q^2}$ and

$$U = -\frac{N\alpha q^2}{4\pi \epsilon_0 R_0} \left(1 - \frac{1}{m}\right)$$

Solution: $U(R) = -N \left[\frac{\alpha q^2}{4\pi \epsilon_0 R} + \frac{B}{R^m} \right] \quad V = V_0 = 2NR_0^3$

Hence, $\frac{dU}{dR} = 0$ at $R = R_0 \Rightarrow R_0^{m-1} = \frac{mB4\pi \epsilon_0}{\alpha q^2} \Rightarrow R_0 = \left(\frac{4\pi \epsilon_0 mB}{\alpha q^2} \right)^{\frac{1}{m-1}}$

$$\frac{1}{K_0} = 2NR_0^3 \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right), \quad \frac{dR}{dV} = \frac{1}{6NR^2} \quad \text{Where } \frac{d^2U}{dR^2} = N \left(-\frac{2\alpha q^2}{4\pi \epsilon_0 R_0^3} + \frac{m(m+1)}{R_0^{m+2}} \right)$$

$$\frac{1}{K_0} = 2NR_0^3 \left(\frac{1}{6NR^2}\right)^2 \cdot N \left(\frac{m(m+1)B}{R_0^{m+2}} - \frac{2\alpha q^2}{4\pi \epsilon_0 R_0^3} \right)$$

$$\frac{1}{K_0} = \frac{1}{18R_0^4} \cdot \left(\frac{m(m+1)B}{R_0^{m-2}} - \frac{2\alpha q^2}{4\pi \epsilon_0} \right) \Rightarrow \frac{1}{K_0} = \frac{1}{18R_0^4} \cdot \left(\frac{m(m+1)B\alpha q^2}{mB4\pi \epsilon_0} - \frac{2\alpha q^2}{4\pi \epsilon_0} \right)$$

$$\Rightarrow \frac{1}{K_0} = \frac{\alpha q^2}{72\pi \epsilon_0 R_0^4} [m+1-2] \Rightarrow m = 1 + \frac{72\pi \epsilon_0 R_0^4}{K_0 \alpha q^2}$$

Total energy, $U(R) = -N \left[\frac{\alpha q^2}{4\pi \epsilon_0 R} + \frac{B}{R^m} \right]$ put $R_0^{m-1} = \frac{mB4\pi \epsilon_0}{\alpha q^2}$ and $U = \frac{-\alpha q^2}{4\pi \epsilon_0 R_0} \left[1 - \frac{1}{m} \right]$

Chapter Three

Crystal Binding

3.4 Some Important Interaction

(a) Covalent Bond

The covalent Bond is the classical electron pair or homopolar bond of chemistry. It is strong bond. It is strong directional property. So carbon, silicon and germanium have the Diamond structure with atoms joined to four nearest neighbours at tetrahedral angle.

Covalent bond is usually formed from two electrons, one from each atom participating in the bond. The electrons forming the bond tend to be partly localised in the region between the two atoms joined by the bond. The spin of the two electrons in the bond are anti parallel. Spin dependent coulomb energy is called exchange interaction.

(b) Metallic bond

In most cases, the outermost electron shell of each of the metal atoms overlaps with a large number of neighbouring atoms. As a consequence, the valence electrons continually move from one atom to another and are not associated with any specific pair of atoms. In short, the

Valence electrons in metals, unlike those in covalently bonded substances, are nonlocalized, capable of wandering relatively freely throughout the entire crystal. The atoms that the electrons leave behind become positive ions, and the interaction between such ions and valence electrons gives rise to the cohesive or binding force that holds the metallic crystal together. Electric conductivity malleability and ductility and other properties of metals can be explained by non-directionality of metallic bond and non-localised nature of electrons.

(c) Hydrogen Bond

Because neutral hydrogen has only one electrons, it should form a covalent bond with only one other atom. It is known under certain condition atom of hydrogen atom attracted by rather stronger force to two atom, thus forming hydrogen bond. It is believed that hydrogen bond is ionic in nature being formed only between the most electro negative atoms. Particularly F, O, and N. in extreme ionic form of the hydrogen bond, the hydrogen atom loses its electron to another atom in molecule ,the bare proton form hydrogen bond.